

**Unit 3 Review**Exponents and Scientific Notation **ANSWER KEY**

Name

Date

Period

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*Show your work. Circle your final answer.*

*HOW TO USE THIS.* This review covers every learning target from the unit — more than the assessment does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

**PART A. Product, Quotient and Power Laws**

Lessons 1–3 · MA.7.NSO.1.1 · 8 questions

1. Write as a single power:  $6^4 \cdot 6^5$

$$6^9$$

*Like bases: add the exponents. The base stays 6.*

2. Write as a single power:  $\frac{3^8}{3^2}$

$$3^6$$

3. Write as a single power:  $(5^2)^4$

$$5^8$$

*Power of a power: multiply the exponents.*

4. Find the value:  $\left(\frac{3}{4}\right)^3$

$$\frac{27}{64}$$

*Power of a quotient: raise the top and the bottom.*

5. Find the value:  $(2 \cdot 5)^3$  and  $2 \cdot 5^3$ . Are they equal?

$$(2 \cdot 5)^3 = 10^3 = 1,000 ; \quad 2 \cdot 5^3 = 2 \cdot 125 = 250 . \text{ No.}$$

*Without parentheses only the 5 is cubed.*

6. Find the value:  $9^0 + 4^0$

$$2$$

*Each zero power is 1, so  $1 + 1 = 2$  — not 0 and not 13.*

7. Find the value of  $n$  :  $\frac{7^n}{7^2} = 7^3$

$$n = 5$$

$$n - 2 = 3.$$

8. Which expression is equivalent to  $4^3 \cdot 4^2$  ?

- ☒ A.  $4^5$   
☐ B.  $16^5$   
☐ C.  $4^6$   
☐ D.  $16^6$

A

## PART B. Zero and Negative Exponents

Lessons 4–5 · MA.8.NSO.1.3 · 7 questions

9. Rewrite with a positive exponent:  $7^{-3}$

$$\frac{1}{7^3}$$

10. Rewrite with a positive exponent:  $\frac{1}{(-5)^{-4}}$

$$(-5)^4$$

The power moves up; the base keeps its parentheses.

11. Find the value:  $3^{-4}$

$$\frac{1}{81}$$

Reciprocal of  $3^4$ . Not  $-81$  and not  $-12$ .

12. Find the value:  $(-4)^{-3}$

$$-\frac{1}{64}$$

1 over  $(-4)^3 = 1$  over  $-64$ . The sign comes from the base.

13. Find the value:  $\left(\frac{2}{5}\right)^{-2}$

$$\frac{25}{4}$$

Flip the fraction, then square it.

14. Find the value:  $\frac{6^0}{6^{-2}}$

$$36$$

$0 - (-2) = 2$ , so  $6^2$ . Or: 1 over  $6^{-2}$  is  $6^2$ .

15. Find the value of  $n$  :  $2^n = \frac{1}{64}$

$$n = -6$$

16. Find the value:  $(2^3)^2 \cdot 2^{-4}$

**4**

$$2^6 \cdot 2^{-4} = 2^2.$$

17. Find the value:  $\frac{5^{-2} \cdot 5^6}{5^3}$

**5**

$$-2 + 6 - 3 = 1.$$

18. **SSDD.** Every part below is about the expression  $4^{-2} \cdot 4^5$ . Same surface, four different questions: rewrite, evaluate, judge an equivalence, solve for an exponent. Only part c has a yes/no answer and only part d has a negative one.

- a. Write it as a single power.

**$4^3$**

$$-2 + 5 = 3.$$

- b. Find its value.

**64**

- c. Is it equivalent to  $\frac{4^5}{4^2}$ ? Explain.

**Yes —  $4^{-2}$  is  $\frac{1}{4^2}$ , so the product is  $\frac{4^5}{4^2} = 4^3$  either way.**

- d. Find  $n$  so that  $4^{-2} \cdot 4^n = 1$ .

**$n = 2$**

$$-2 + n = 0.$$

19. Select ALL of the expressions equivalent to  $3^{-2}$ .

☐ A.  $-6$

☐ B.  $-9$

☒ C.  $3^2 \cdot 3^{-4}$

☒ D.  $\frac{3^4}{3^6}$

☒ E.  $\frac{1}{9}$

☒ F.  $\left(\frac{1}{3}\right)^2$

**C, D, E and F**

20. Aryella says that  $10^0$  and  $134^0$  are equal. Is she right? Explain.

**Yes. Any nonzero base to the zero power is 1, so both equal 1.**

21. Write as a single monomial:  $x^7 \cdot x^5$   
 $x^{12}$

22. Write as a single monomial:  $(2m^4)^3$   
 $8m^{12}$   
*The 2 is cubed too.*

23. Write as a single monomial:  $(3a^2b)(5ab^4)$   
 $15a^3b^5$

24. Write as a single monomial:  $\frac{18x^6y^3}{6x^2y^3}$   
 $3x^4$   
 *$y^3$  over  $y^3$  is  $y^0 = 1$ .*

25. Rewrite with positive exponents only:  $4x^{-3}$   
 $\frac{4}{x^3}$   
*Only the  $x$  moves.*

26. Rewrite with positive exponents only:  $(2y^{-2})^3$   
 $\frac{8}{y^6}$

27. Rewrite with positive exponents only:  $\frac{a^3}{a^8}$   
 $\frac{1}{a^5}$   
 $3 - 8 = -5$ .

28. Which expression is equivalent to  $(5x^2)^3$  ?

- ☐ A.  $5x^6$   
☒ B.  $125x^6$   
☐ C.  $125x^5$   
☐ D.  $15x^6$

**B**

**PART E. Large Numbers in Scientific Notation**

Lesson 8 · MA.8.NSO.1.4 · 5 questions

29. Write 7,400,000 in scientific notation.

$$7.4 \times 10^6$$

30. Write 205,000,000,000 in scientific notation.

$$2.05 \times 10^{11}$$

*The interior zero stays in the coefficient.*

31. Write  $8.3 \times 10^5$  in standard form.

$$830,000$$

32. How many times larger is  $9 \times 10^7$  than  $3 \times 10^4$  ?

$$3 \times 10^3 = 3,000 \text{ times}$$

$$9 \div 3 = 3 \text{ and } 10^7 \div 10^4 = 10^3.$$

33. Dana writes 45,000 as  $45 \times 10^3$ . Is her expression equal to 45,000? Is it scientific notation? Explain.

**Equal, yes. Scientific notation, no — 45 is not less than 10. The correct form is  $4.5 \times 10^4$ .**

**PART F. Small Numbers, and How Many Times**

Lesson 9 · MA.8.NSO.1.4 · 6 questions

34. Write  $10^{-5}$  as a decimal.

$$0.00001$$

*A 1 in the fifth decimal place.*

35. Write 0.00093 in scientific notation.

$$9.3 \times 10^{-4}$$

*The 9 is in the fourth decimal place.*

36. Write  $6.1 \times 10^{-7}$  in standard form.

$$0.00000061$$

37. Which is greater,  $4.8 \times 10^{-3}$  or  $9.1 \times 10^{-4}$  ? Explain.

$$4.8 \times 10^{-3} \text{ — the exponent } -3 \text{ is greater than } -4; 0.0048 > 0.00091.$$

38. How many times larger is  $8.4 \times 10^{-2}$  than  $2.1 \times 10^{-6}$  ?

$$4 \times 10^4 = 40,000 \text{ times}$$

$$8.4 \div 2.1 = 4 \text{ and } -2 - (-6) = 4.$$

- 39.** A computer displays a number as  $2.7\text{E}-5$ . Write it in scientific notation and in standard form.

$$2.7 \times 10^{-5} = 0.000027$$